Project Work: Operating System and System Programming

Title: Installation of FreeBSD Operating System in Virtual Environment (VMware Workstation)

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Course: Operating System and System Programming (OSSP)
Type: Individual Assignment
Operating System Installed: FreeBSD 10 (64-bit)
Virtualization Tool: VMware Workstation

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A. Introduction (Background and Motivation)

FreeBSD is a UNIX-like operating system that originated from the Berkeley Software Distribution (BSD) developed at the University of California, Berkeley in the 1970s. BSD evolved from the original UNIX operating system created at Bell Laboratories. FreeBSD was officially released in 1993 as a free and open-source operating system designed for stability, networking performance, and server reliability.

FreeBSD is widely used in enterprise servers, network infrastructure, cybersecurity systems, and high-performance computing environments. Major technology companies

and internet service providers use FreeBSD because of its advanced networking stack, strong security model, and reliable filesystem support such as the ZFS filesystem.

The motivation for this project is to gain practical experience in installing and configuring an operating system in a virtual environment using VMware Workstation. Virtualization allows operating systems to be tested, configured, and maintained without affecting the physical computer system.

B. Objectives

The objectives of this project are:

- To install the FreeBSD operating system in a virtual environment using VMware Workstation
- To understand the historical development of the FreeBSD operating system
- To configure system settings including user accounts
- To identify and resolve technical problems encountered during installation
- To understand filesystem support used in modern operating systems
- To demonstrate knowledge of virtualization in modern operating systems

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C. Requirements

The system requirements for installing FreeBSD in a virtual environment are categorized into Minimum, Recommended, and Used configurations to clearly show the practical environment where the installation was performed.

(i) Hardware Requirements

Minimum Hardware Requirements:

- Processor: 64-bit CPU (Intel or AMD) with virtualization support
- RAM: 1 GB
- Storage: 20 GB free disk space
- Network: Ethernet or Wi-Fi adapter

Recommended Hardware Requirements:

- Processor: Dual-core 64-bit CPU (Intel Core i3/i5/i7 or AMD equivalent)
- RAM: 2 GB or higher
- Storage: 40 GB or higher virtual disk
- Network: Stable internet connection for updates and package installation

Hardware Used in This Project:

- Processor: Intel Core i5 (64-bit) processor
- RAM: 16 GB system memory
- Storage: 512 GB, and 20 GB virtual disk allocated to the virtual machine
- Host Machine: Laptop running Windows 10 operating system
- Network: Wireless network adapter

(ii) Software Requirements

Minimum Software Requirements:

- Virtualization Software: VMware Workstation or Oracle VM VirtualBox
- Operating System Image: FreeBSD ISO file
- Host Operating System: Windows, Linux, or macOS

Recommended Software Requirements:

- Virtualization Software: VMware Workstation Pro 16 or later
- Operating System Image: Stable FreeBSD release version
- Host Operating System: Windows 10 or later with virtualization enabled in BIOS/UEFI

Software Used in This Project:

- Virtualization Software: VMware Workstation Pro
- Guest Operating System: FreeBSD 10 (64-bit)
- Host Operating System: Microsoft Windows 11
- Installation Media: FreeBSD ISO image file

D. Installation Steps

Step 1: Download and Launch VMware Workstation

The VMware Workstation installation software was downloaded from the official distribution source available on the TechSpot website. The version downloaded was compatible with the Windows operating system installed on the host computer.

After completing the download process, VMware Workstation was installed and launched successfully. Once the application started, the **"Create a New Virtual Machine"** option was selected from the main interface to begin the virtual machine configuration process for installing the FreeBSD operating system.

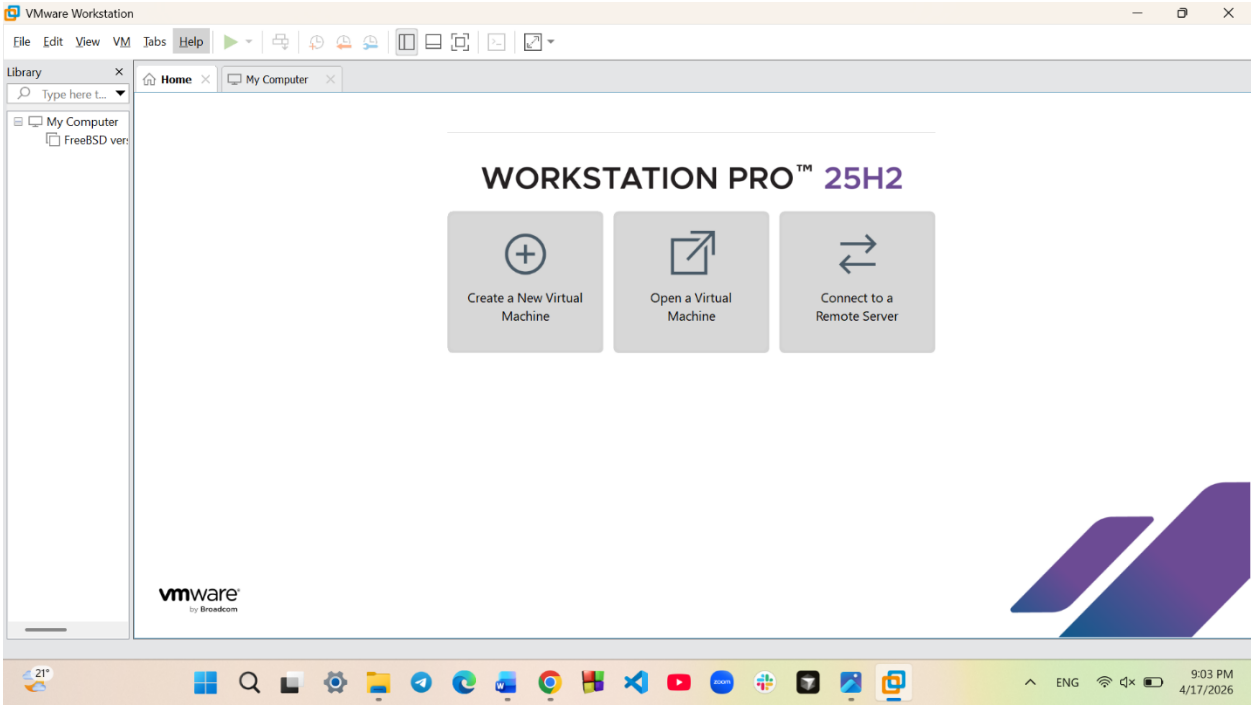

Figure 1: VMware Workstation main interface showing the option to create a new virtual machine.

Step 2: Create a New Virtual Machine

After launching VMware Workstation, the option to create a new virtual machine was selected from the main interface. The **Custom (Advanced)** configuration method was chosen to allow manual control over the virtual machine parameters.

During the virtual machine setup process, the following configuration settings were defined:

- **Virtual Machine Name:** FreeBSD_VM
- **Guest Operating System Type:** FreeBSD
- **Version:** FreeBSD 64-bit
- **Memory Allocation:** 4 GB (4096 MB) RAM assigned to the virtual machine
- **Processor Configuration:** 1 processor with 2 cores
- **Virtual Disk Type:** Virtual disk created as a new disk
- **Virtual Disk Size:** 20 GB (SCSI virtual disk) allocated for the FreeBSD installation
- **Network Type:** Network Address Translation (NAT) selected to enable internet connectivity

These configuration parameters ensured that the virtual machine had sufficient system resources to support the installation and operation of the FreeBSD operating system in a stable virtualized environment.

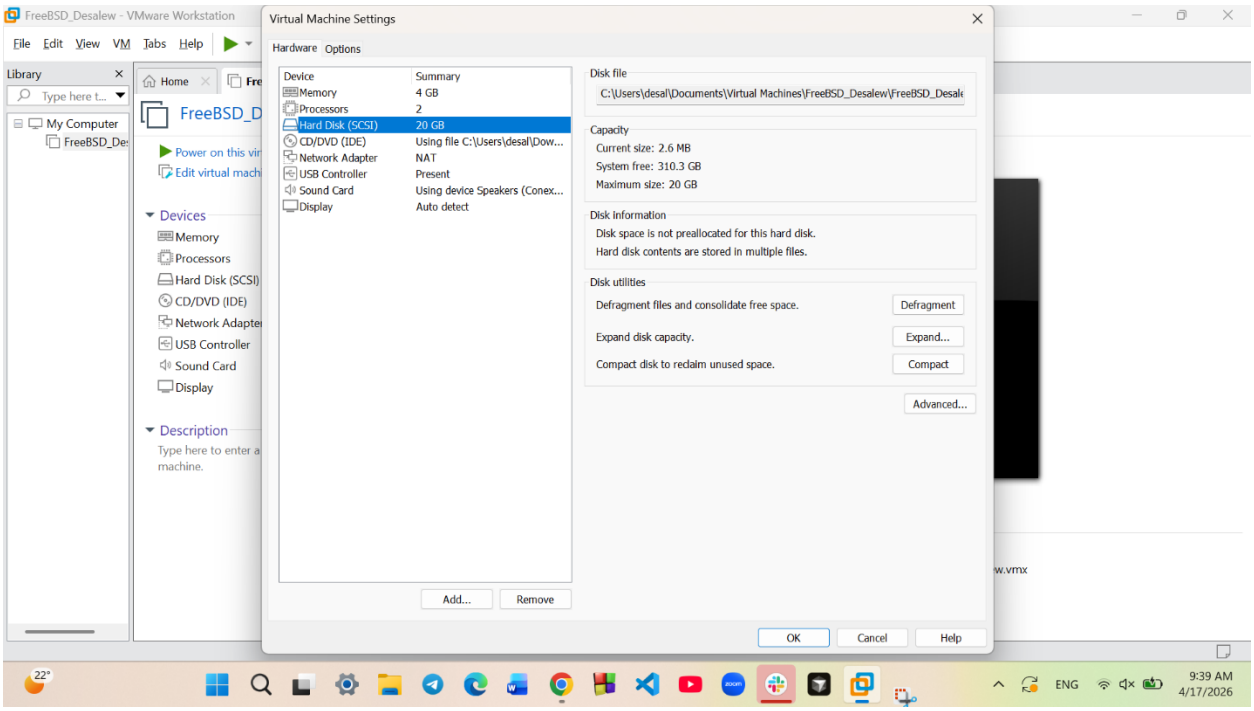

Figure 2: Virtual machine configuration settings showing memory allocation, processor configuration, and virtual disk size.

Step 3: Configure Installation Media (ISO Image File)

After creating the virtual machine, the installation media for the FreeBSD operating system was configured using the Virtual Machine Settings interface in VMware Workstation.

The **CD/DVD (IDE)** device option was selected from the hardware configuration list. The **Use ISO image file** option was enabled to allow the virtual machine to boot from an installation image. The FreeBSD installation file (FreeBSD-15.0-RELEASE-amd64.iso) was then selected from the local storage directory.

- **CD/DVD Device:** ISO image file selected for FreeBSD installation

These settings ensured that the virtual machine was properly configured to boot and begin the installation process of the FreeBSD operating system.

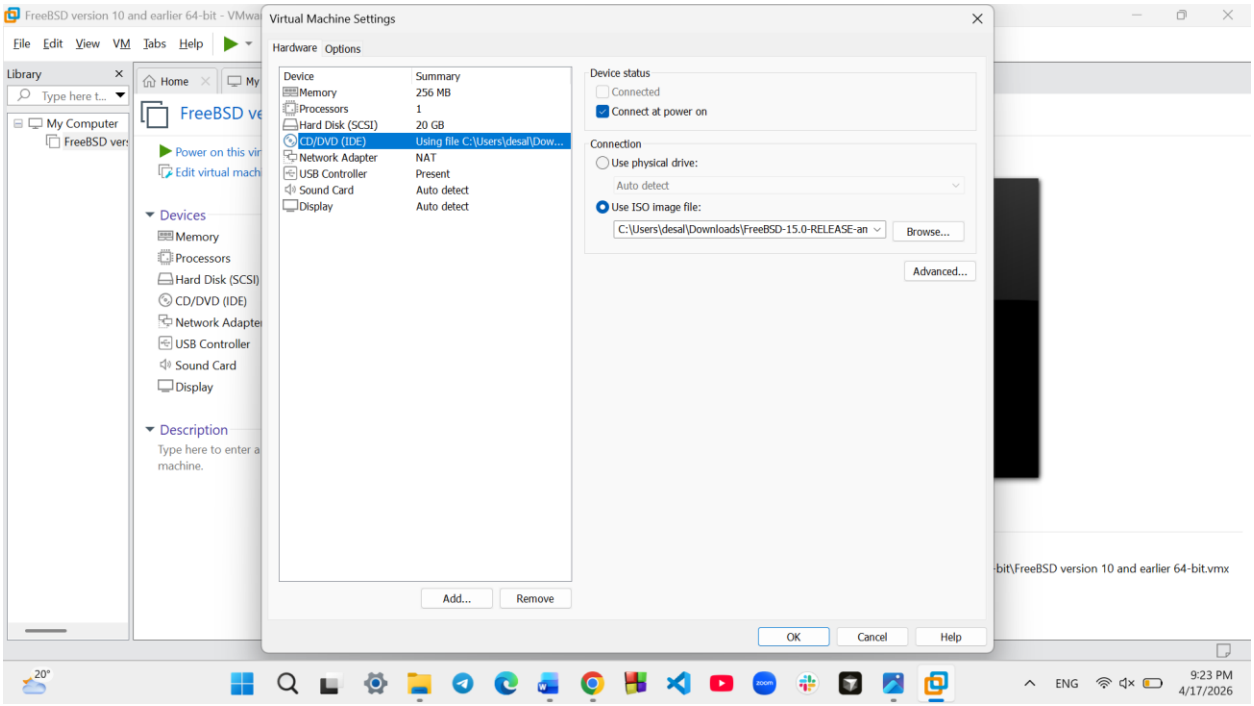

Figure 3: Virtual Machine Settings window showing the selected ISO image file

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Step 4: Start the Virtual Machine

After completing the virtual machine configuration and attaching the installation media, the virtual machine was powered on to initiate the operating system installation process in VMware Workstation.

The **Power on this virtual machine** option was selected from the main interface. As a result, the system booted from the attached FreeBSD ISO image file, and the FreeBSD installer loaded successfully. The initial installation screen displayed the available boot options and system initialization messages, confirming that the virtual machine was correctly configured to begin the installation procedure.

This step marked the start of the FreeBSD operating system installation within the virtual environment.

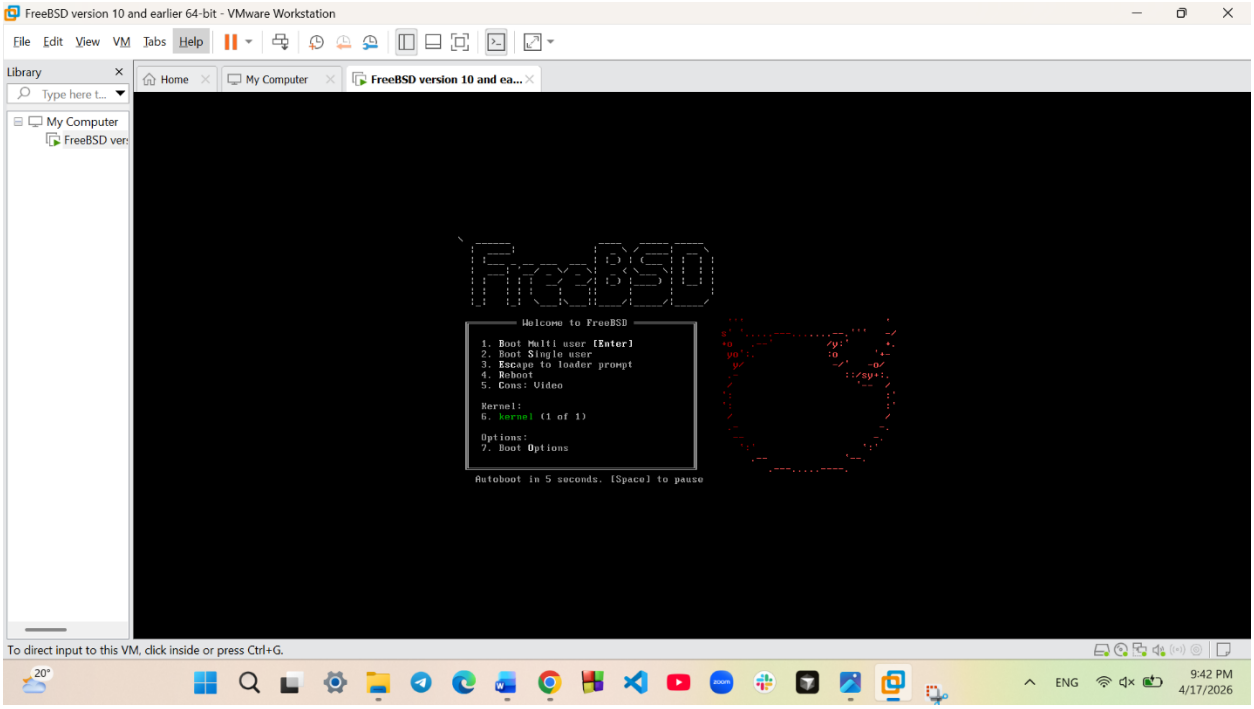


Figure 4: FreeBSD installation start screen showing the system booting from the installation media.

Step 5: Configure Disk Partition and Filesystem

During the installation process of the FreeBSD operating system, disk partitioning and filesystem configuration were performed to prepare the virtual storage device for system installation.

The installer prompted for disk configuration options, and the **Zettabyte File System (ZFS)** was selected as the primary filesystem for the virtual disk. ZFS was chosen because it provides advanced features such as data integrity verification, automatic error detection, snapshot capability, and high reliability in managing large storage volumes.

The virtual disk was automatically partitioned by the installer using the default ZFS layout, which created the necessary system partitions required for the FreeBSD operating system to function properly. This configuration ensured efficient storage management and improved system stability within the virtual environment.

The use of the **ZFS** filesystem is widely recommended in modern FreeBSD installations due to its robustness, scalability, and built-in data protection mechanisms.

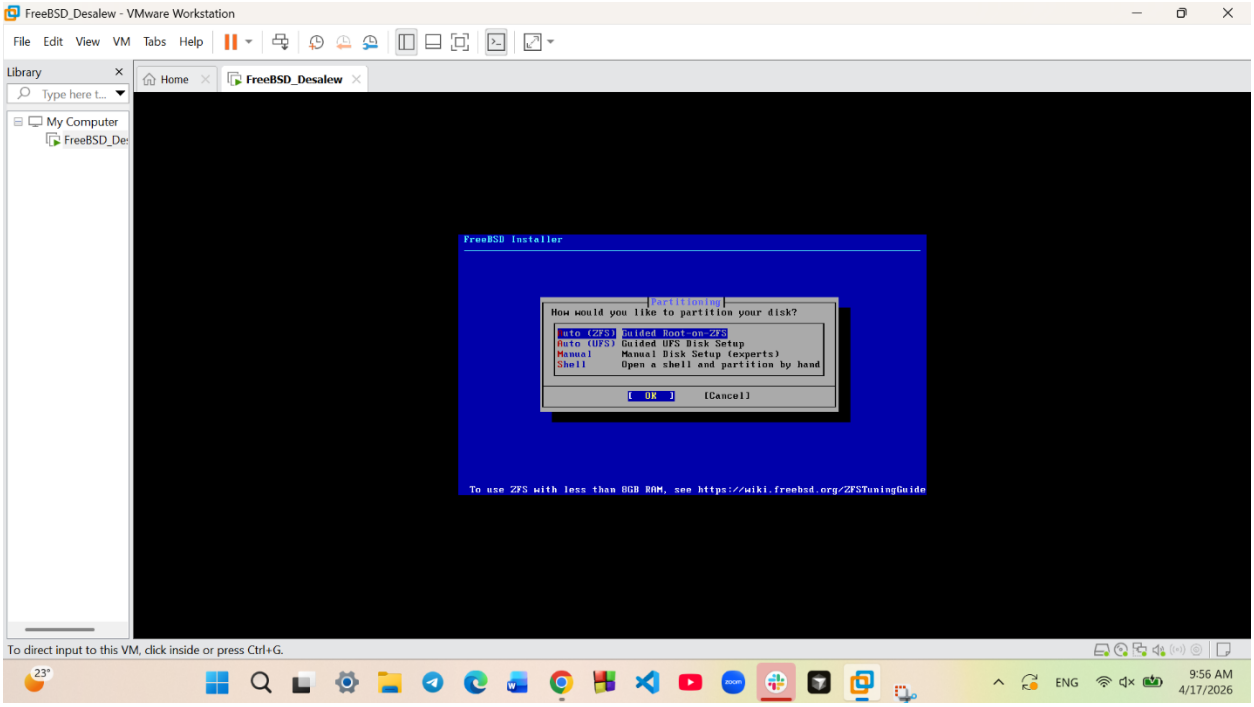


Figure 5: Disk partition and ZFS filesystem configuration screen showing the selected filesystem and disk layout.

Step 6: Create User Account Using Full Name

During the installation process of the FreeBSD operating system, a standard user account was created to enable secure system access and routine administrative operations.

The installer prompted for user account details, including the full name, username, and password. In accordance with the assignment requirement, the user account was created using the full name **"Desalew Gashaye"**. The corresponding username was defined for login purposes, and a secure password was configured to protect system access.

Creating a user account with the full name ensures proper identification of the system user and supports accountability in multi-user operating system environments. This step also

allows the user to perform normal system operations without relying solely on the root administrator account, thereby improving system security and management.

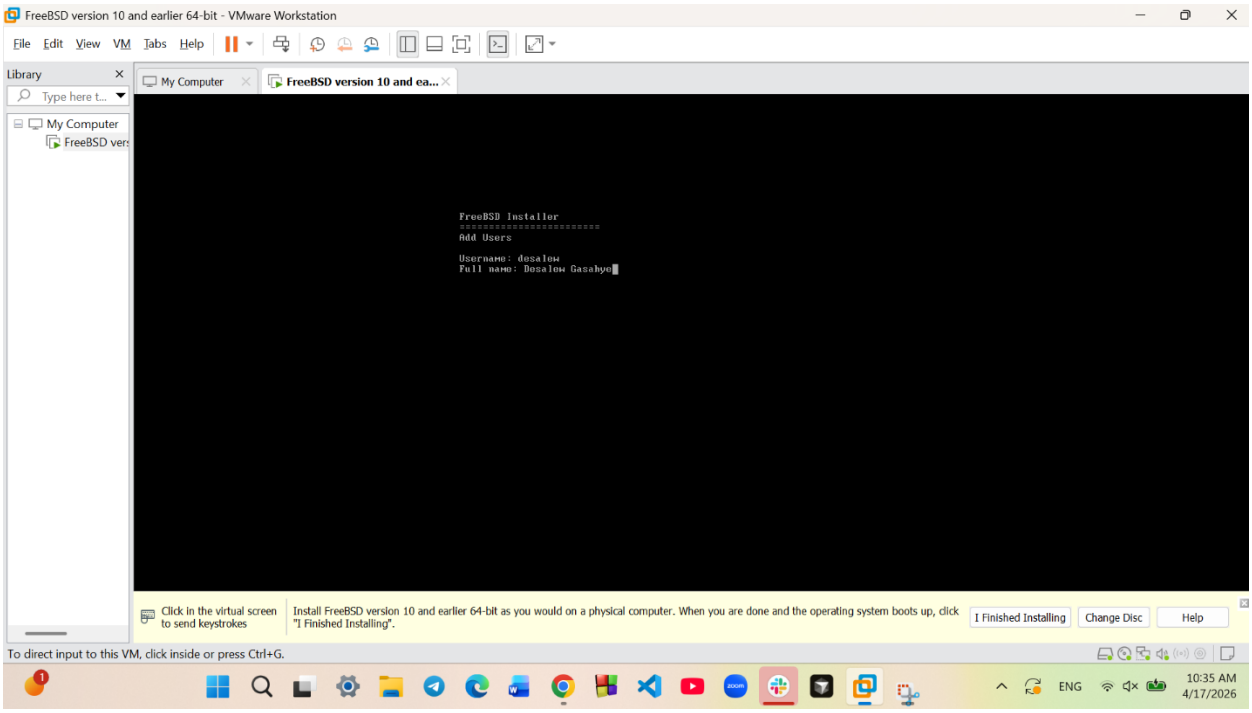


Figure 6: User account creation screen showing the full name **"Desalew Gashaye"** entered during the FreeBSD installation process.

Step 7: System Login

After completing the installation and system configuration, the virtual machine was restarted to load the installed FreeBSD operating system. Upon startup, the system displayed the login interface, prompting for user authentication credentials.

The system login was performed using the previously created username and password associated with the user account **"Desalew Gashaye"**. Successful authentication confirmed that the operating system had been installed correctly and that the user account configuration was functioning as expected.

This step verified that the FreeBSD operating system was operational within the virtual environment provided by VMware Workstation and ready for normal system usage.

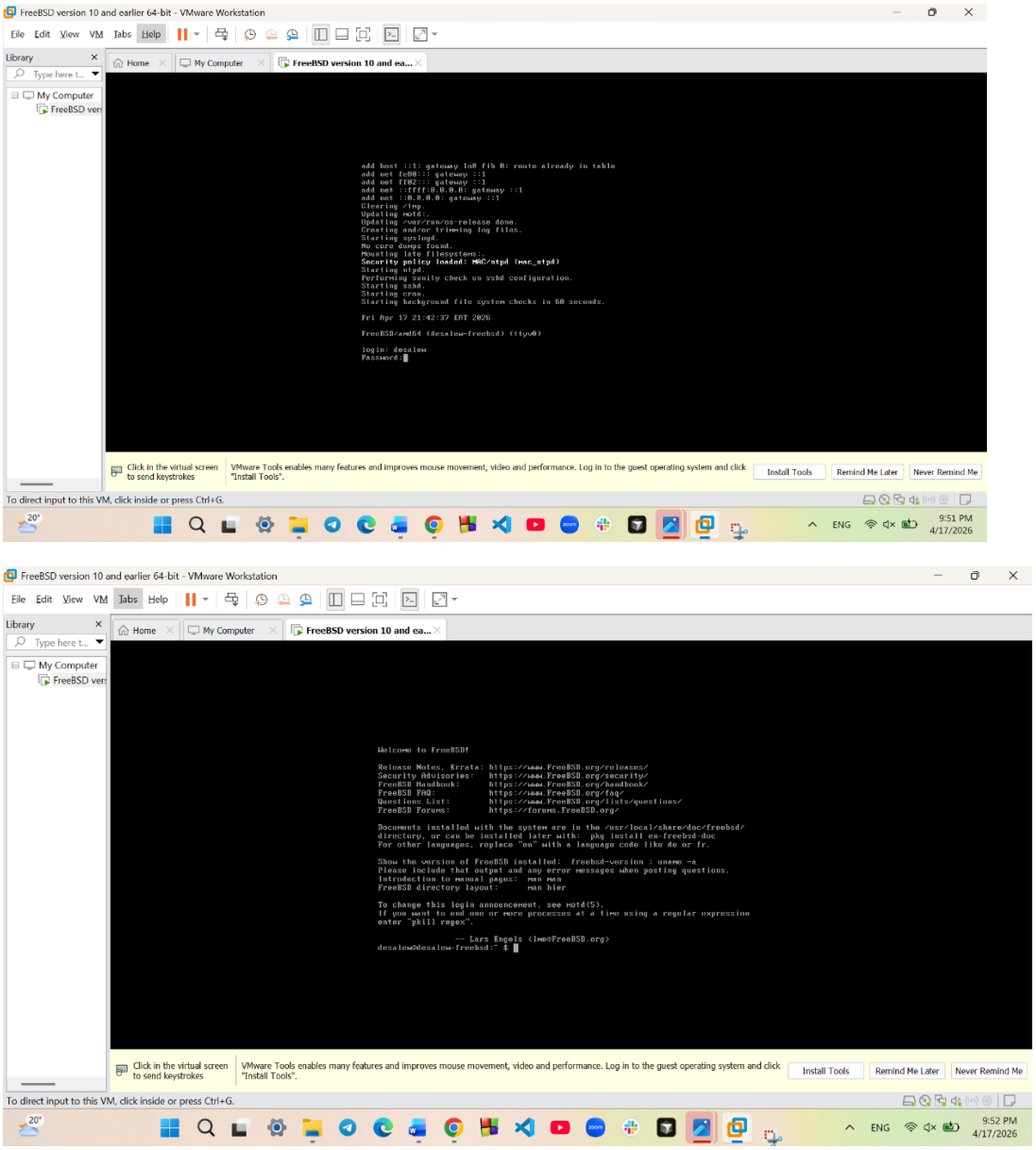


Figure 7: System login screen showing successful user authentication into the FreeBSD operating system.

Step 8: Switch to Root User

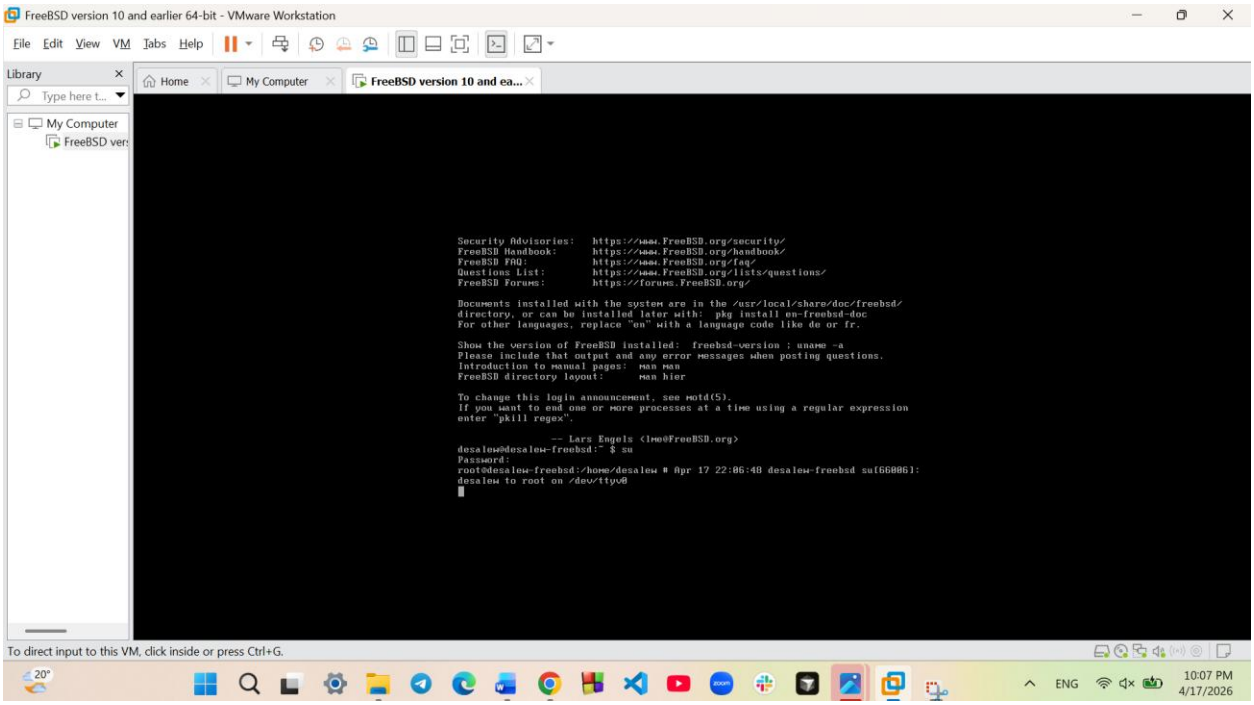
After successfully logging into the FreeBSD system using the standard user account, administrative privileges were required to perform system-level operations and configuration tasks. To obtain administrator access, the system command below was executed in the terminal environment.

su

The su (substitute user) command was used to switch from the standard user account to the **root** user account, which has full system administration privileges in the FreeBSD operating system. Upon entering the correct root password, the system granted elevated privileges, allowing the execution of administrative commands and system configuration activities.

Successful access to the root account confirmed that the administrator credentials were correctly configured during the installation process and that the system security authentication mechanism was functioning properly.

This step is essential in UNIX-based operating systems because the root user is responsible for managing system resources, installing software packages, and performing maintenance operations.



8: Term inal window showing successful root login after executing the su command.

Step 9: Configure System Date and Time

After successfully accessing the system with administrative privileges, the system date and time were configured to ensure accurate system logging, file timestamp management, and proper synchronization of system processes.

The date and time were set using the system administration command in the terminal environment of the FreeBSD operating system. The command was executed with the correct numerical format representing the year, month, day, hour, and minute.

date 202604171121

This command configured the system date to **April 17, 2026**, and the system time to **11:21 AM**. Proper configuration of system time is essential in operating systems because it supports accurate event recording, authentication processes, and network communication.

Successful execution of the command confirmed that the system clock was correctly updated and that administrative control of system configuration was functioning as expected.

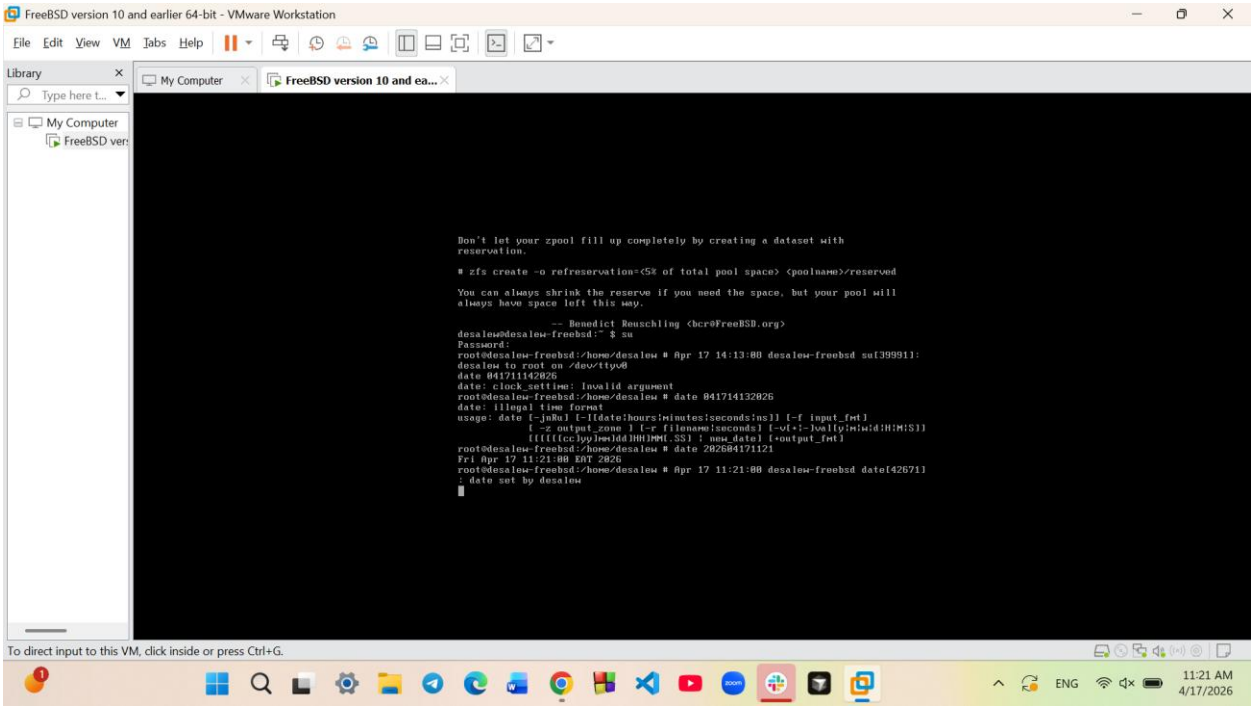


Figure 9: Terminal window showing successful execution of the date 202604171121 command and confirmation of the updated system date and time.

E. Issues or Problems Faced

Problem 1: Pager Read Error

During the installation process of the FreeBSD operating system in the virtual environment, the system displayed a **pager read error** message after the virtual machine was unintentionally disconnected before the installation reboot process was completed.

This interruption prevented the operating system from properly loading system files required during the boot sequence. As a result, the system was unable to continue the normal startup procedure and displayed an error message indicating a failure to read data from the virtual disk.

The issue occurred while running the operating system inside VMware Workstation, where the virtual machine session was terminated before the installation process had finalized all system configuration tasks.

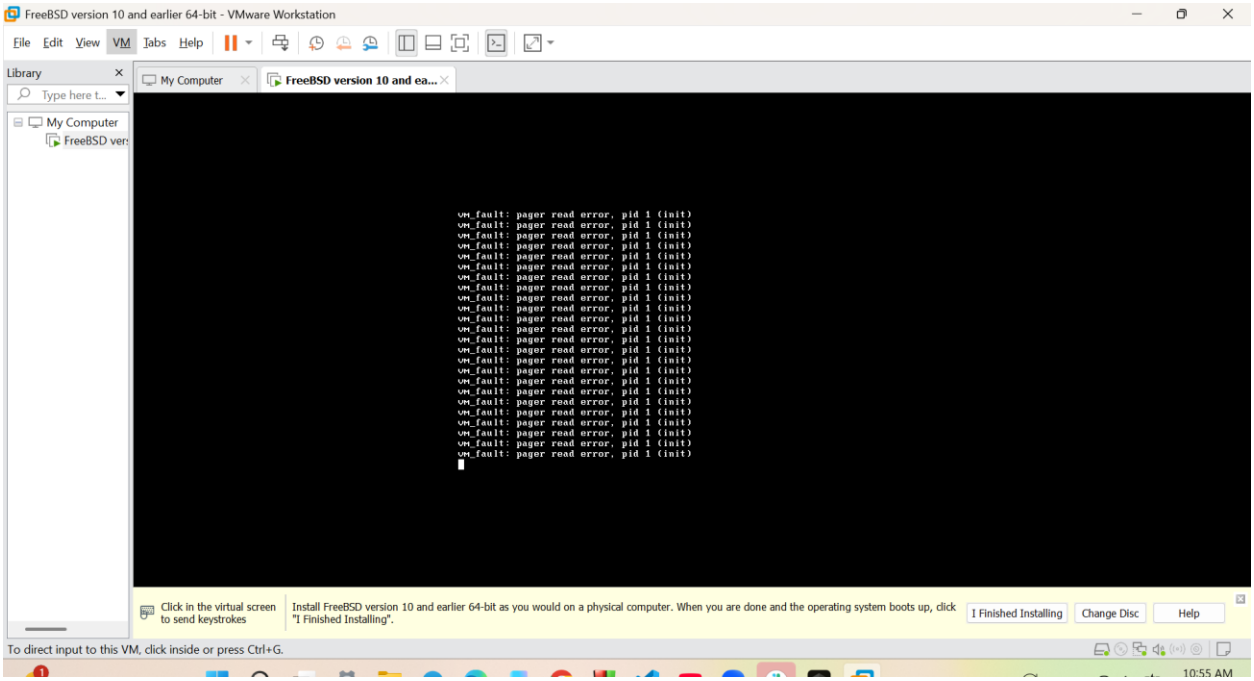


Figure 10: Pager read error message displayed during system startup.

F. Solutions

Solution to Problem 1

Restart the virtual machine and allow the operating system to complete the reboot process normally.

G. Filesystem Support

Supported Filesystems and Their Usage

NTFS – Used in Microsoft Windows operating systems because it supports file permissions, encryption, and large file storage.

FAT32 – Used in USB flash drives and memory cards because it is compatible with multiple operating systems.

exFAT – Used in modern removable storage devices because it supports large files and cross-platform compatibility.

ext4 – Used in Linux operating systems because it provides journaling, performance, and reliability.

Btrfs – Used in advanced Linux systems because it supports snapshots and data integrity features.

HFS+ – Used in older macOS systems because it manages file storage efficiently on Apple devices.

APFS – Used in modern macOS systems because it supports encryption, fast performance, and optimized storage management.

ZFS – Used in FreeBSD because it provides data integrity verification, automatic error correction, and snapshot capability.

H. Advantages and Disadvantages

Advantages of FreeBSD

- High system stability and reliability
- Strong network performance and security
- Advanced filesystem support using ZFS
- Open-source operating system with community support

Disadvantages of FreeBSD

- Limited commercial software availability compared to Windows

- Requires technical knowledge for system administration
- Smaller hardware driver support compared to Linux distributions

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I. Conclusion

The FreeBSD operating system was successfully installed in a virtual environment using VMware Workstation. The installation process demonstrated practical skills in virtualization, system configuration, filesystem management, and troubleshooting. The project provided hands-on experience with UNIX-based operating systems and system administration commands.

J. Future Outlook / Recommendation

Future work may include:

- Configuring network services in FreeBSD
- Implementing system security and firewall configuration
- Practicing shell scripting and system automation
- Deploying FreeBSD in server environments
- Exploring virtualization and cloud computing technologies

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2. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a technology that allows multiple operating systems to run on a single physical computer by creating virtual machines that simulate hardware resources such as CPU, memory, storage, and network interfaces.

Why Virtualization is Used

Virtualization is used to improve hardware utilization, reduce operational costs, provide system isolation for security, and allow testing of operating systems without affecting the main computer system.

How Virtualization Works

Virtualization works through a software component called a hypervisor. The hypervisor manages physical hardware resources and allocates them to virtual machines. VMware Workstation acts as the hypervisor in this project by providing virtual hardware that allows FreeBSD to run as a guest operating system inside a Windows host operating system.